Map Reduce
Design Patterns

Caveat

- The great power of MapReduce lays in its simplicity
  - The programmer only needs to prepare input data, configuration information, implement mapper and reducer and, optionally, setup, cleanup, combiner and partitioner
  - All other aspects are handled by the framework
- This simplicity has some pros and cons
  - Pros -- It provides clear boundaries for what you can do and cannot do
  - Cons -- The programmer has to express any algorithm in terms of a small number of rigidly defined components that must fit together in very specific ways
- How to <u>recast/reshape</u> an algorithm into this programming model, i.e., as a sequence of one or more MapReduce jobs?
- Good programmers can produce bad MapReduce algorithms
- MapReduce design patterns
  - Principles/guidelines to build better software
  - Not specific to a domain (e.g., text processing). Instead, it is a general approach to solve a problem

Design Patterns

- In-mapper Combining
- Pairs and stripes
- Patterns for relational algebra operators
- Patterns for matrix-vector and matrix-matrix multiplication

In-mapper Combining (I)

Intermediate data (i.e., mappers outputs) are

- Written locally
- Transferred from mappers to reducers over cluster network

- Issue
  - Bottleneck: Writing to disk and transferring over the network are relatively «expensive» operations
  - Reducing intermediate data translates into increasing algorithmic efficiency
- Solution: Local aggregation
  - Use combiners
  - Use In-Mapper Combining

In-Mapper Combining (II)

Pseudo-code of the basic word-count algorithm

```
class Mapper
    method Map(docid a, doc d)
        for all term t ∈ doc d do
            Emit(term t, count 1)

class Reducer
    method Reduce(term t, counts [c1, c2, . . .])
        sum ← 0
        for all count c ∈ counts [c1, c2, . . .] do
            sum ← sum + c
        Emit(term t, count sum)
```

In-Mapper Combining (III)

Pseudo-code of stateless in-mapper combining

```
class Mapper
    method Map(docid a, doc d)
        H ← new AssociativeArray
        for all term t ∈ doc d do
            H{t} ← H{t} + 1
        for all term t ∈ H do
            Emit(term t, count H{t})
```

In-Mapper Combining (IV)

Pseudo-code of stateful in-mapper combining (the true in-mapper combining)

```
class Mapper
    method Initialize
        H ← new AssociativeArray
    method Map(docid a, doc d)
        for all term t ∈ doc d do
            H{t} ← H{t} + 1
    method Close
        for all term t ∈ H do
            Emit(term t, count H{t})
```

In-Mapper Combining (V)

- Advantages:
  - Combiners are provided as an optimization to the execution framework, which has the option of using it (perhaps multiple times) or not at all. On the other hand, in-mapper combining provides complete control on local aggregation process (how and when)
  - Combiners only reduce the amount of intermediate data that is transferred across the network but do not reduce intermediate data emitted by mappers in the first place. In-mapper combining provides further control by avoiding unnecessary object creation as output to the mapper
- Disadvantages:
  - Potential ordering-dependent bugs – preserving state across records may cause algorithmic behavior to depend on the order in which input records are encountered
  - Memory scalability bottleneck -- the associative array may no longer fit in memory. Solved by:
    - Either emitting intermediate data after processing every *n* (how big?) key-value pairs rather than all of them
    - Or monitoring memory footprint and flush intermediate data to disk once memory usage has exceeded a certain (how big?) threshold

Matrix Generation

- Common problem:
  - Given an input of size N, generate a squared output of size $N \times N$
- Example: word co-occurrence matrix
  - Given a document collection, emit the bigram frequencies
  - Where N is the number of unique words
  - Cell m_{ij} contains the number of times word w_i co-occurs with word w_j
  - Used for example in statistical natural language processing
- Two solutions
  - Pairs: generating $O(N^2)$ data in $O(1)$ space
  - Stripes: generation $O(N)$ data in $O(N)$ space

Pairs

class MAPPER
    method MAP(docid a, doc d)
        for all term $w \in$ doc d **do**
            for all term $u \in$ NEIGHBORS(w) **do**
                EMIT(pair (w, u), count 1) ▷ Emit count for each co-occurrence

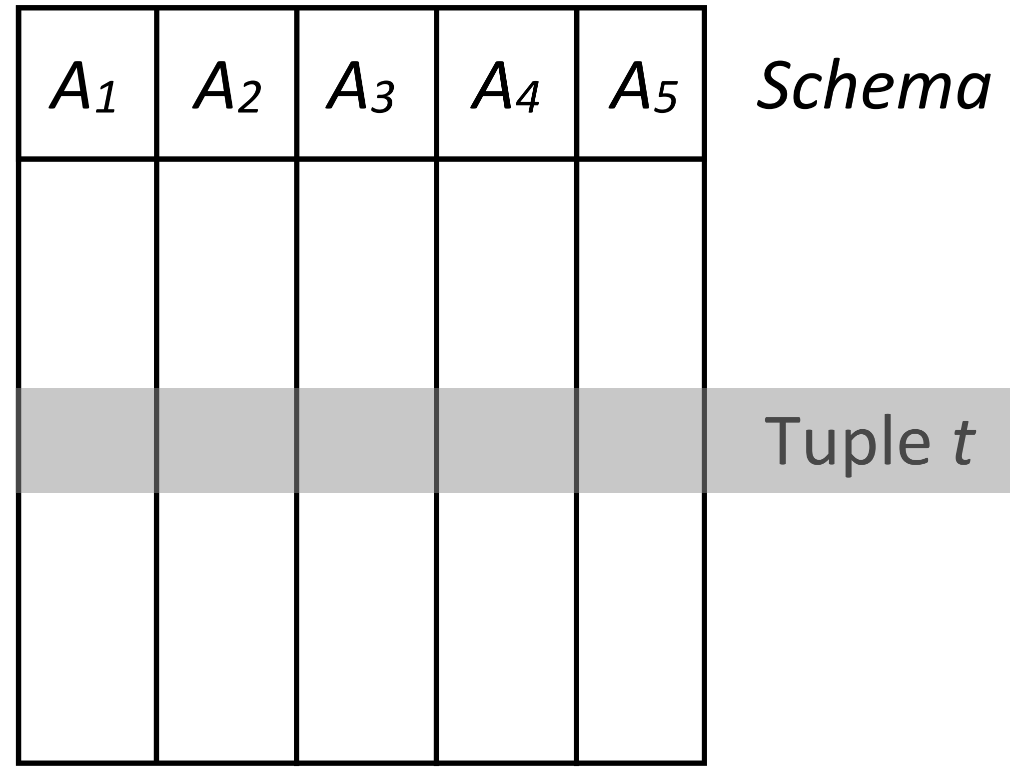
class REDUCER
    method REDUCE(pair p, counts $[c_1, c_2, \ldots]$)
        $s \leftarrow 0$
        for all count $c \in$ counts $[c_1, c_2, \ldots]$ **do**
            $s \leftarrow s + c$ ▷ Sum co-occurrence counts
        EMIT(pair p, count s)

Stripes

```
class Mapper
    method Map(docid a, doc d)
        for all term w ∈ doc d do
            H ← new AssociativeArray
            for all term u ∈ Neighbors(w) do
                H{u} ← H{u} + 1                    ▷ Tally words co-occurring with w
            Emit(Term w, Stripe H)

class Reducer
    method Reduce(term w, stripes [H1, H2, H3, . . .])
        Hf ← new AssociativeArray
        for all stripe H ∈ stripes [H1, H2, H3, . . .] do
            Sum(Hf, H)                             ▷ Element-wise sum
        Emit(term w, stripe Hf)
```

Relational Algebra Operators



Relation *R*

- SELECTION: Select from relation *R* tuples satisfying condition *c(t)*
- PROJECTION: For each tuple in relation *R*, select only certain attributes A_i
- UNION, INTERSECTION, DIFFERENCE: Set operations on two relations with same schema
- NATURAL JOIN
- GROUPING and AGGREGATION

Selection and projection

Selection

- Map: each tuple t in R, if condition $c(t)$ is satisfied, is outputted as a $(t, \perp)$ pair
- Reduce: for each $(t, \perp, \perp, \perp, \ldots)$ pair in input, output $(t, \perp)$

Projection

- Map: each tuple t in R, create a new tuple t' containing only the projected attributes and output a $(t', \perp)$ pair
- Reduce: for each $(t', \perp, \perp, \perp, \ldots)$ pair in input, output $(t', \perp)$

Union, intersection and difference

Union

- Map: for each tuple t in R and S, output a $(t, \perp)$ pair
- Reduce: for each input key t, there will be 1 or 2 values equal to $\perp$. Coalesce them in a single output $(t, \perp)$

Intersection

- Map: for each tuple t in R and S, output a $(t, \perp)$ pair
- Reduce: for each input key t, there will be 1 or 2 values equal to $\perp$. If there are 2 value, coalesce them in a single output $(t, \perp)$, otherwise do nothing

Difference

- Map: for each tuple t in R, output (t , R) and for each tuple t in S, output (t , S)
- Reduce: for each input key t, there will be 1 or 2 values. If there is 1 value equal to (t , R) output $(t, \perp)$, otherwise do nothing

Natural Join

For simplicity, assume we have two relations *R(A,B)* and *S(B,C)*. Find tuples that agree on the *B* attribute values and output them.

Natural join

- Map: for each tuple *(a, b)* from *R*, output *(b, (*R*, a))* and for each tuple *(b, c)* from *S*, produce *(b, (*S*, c))*
- Reduce: For each input key *b*, there will be a list of values of the form *(*R*, a)* or *(*S*, c)*. Construct all pairs and output them together with *b*

Grouping and aggregation

For simplicity, assume we have the relation *R(A,B,C)* and we want to group-by *A* and aggregate on *B,* disregarding C.

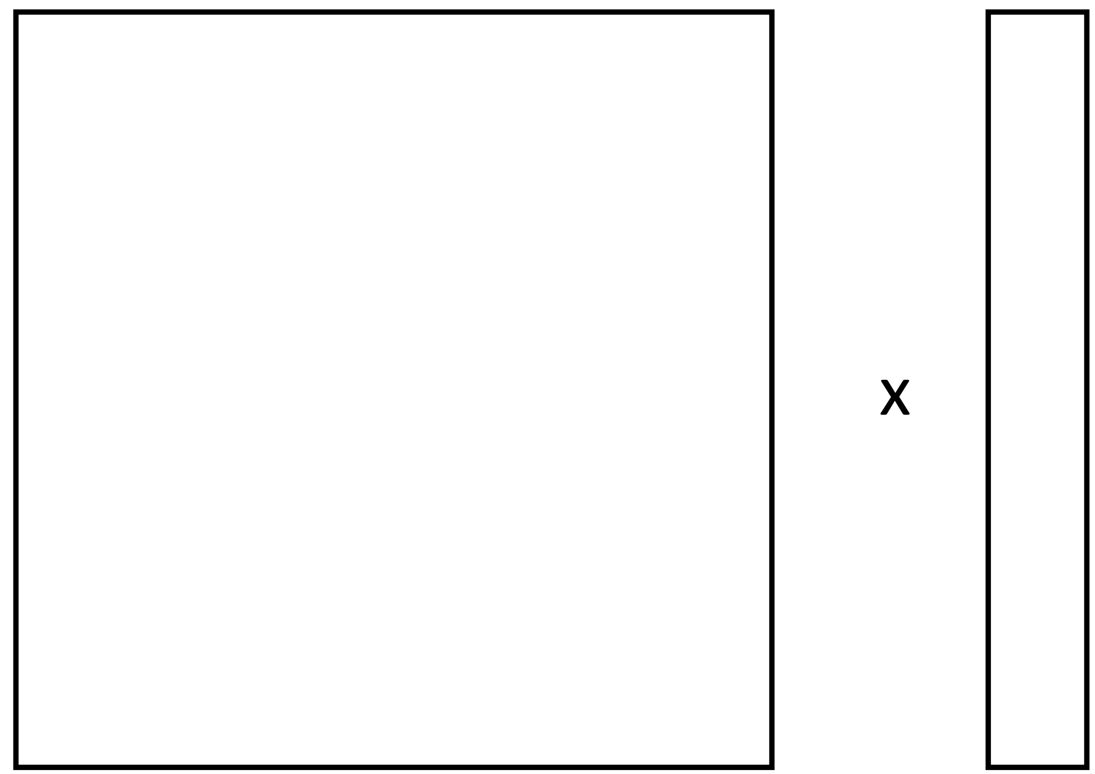
Grouping and aggregation

- Map: for each tuple *(a, b, c)* from *R*, output *(a, b)*. Each key *a* represents a group.
- Reduce: apply the aggregation operator to the list of *b* values associated with group keyed by *a*, producing *x*. Then output *(a, x).*

Stage Chaining

- As map reduce calculations get more complex, it’s useful to break them down into stages, with the output of one stage serving as input to the next

Matrix-Vector Multiplication



Matrix M
size m x n

Vector v
size n x 1

The matrix does not fit in memory, and

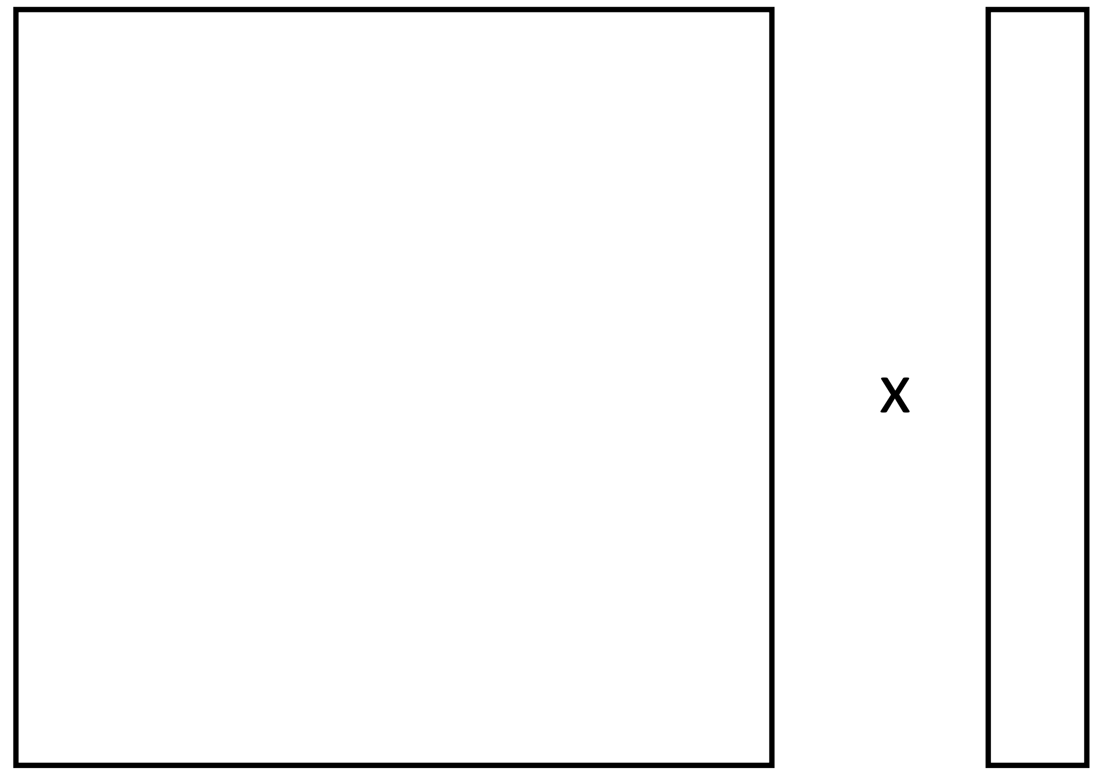
1. The vector v does fit in a machine's memory
2. The vector v does not fit in machine's memory

A graphical and simple example of Matrix Vector Multiplication

$$\begin{bmatrix} 1 & 5 \\ 2 & 4 \\ 3 & 6 \end{bmatrix} \times \begin{bmatrix} 7 \\ 2 \end{bmatrix} = \begin{bmatrix} 17 \\ 22 \\ 33 \end{bmatrix}$$

Matrix *M*	Vector *v*	Vector *r*
size *3* x *2*	size *2* x *1*	size *3* x *1*

Vector does fit



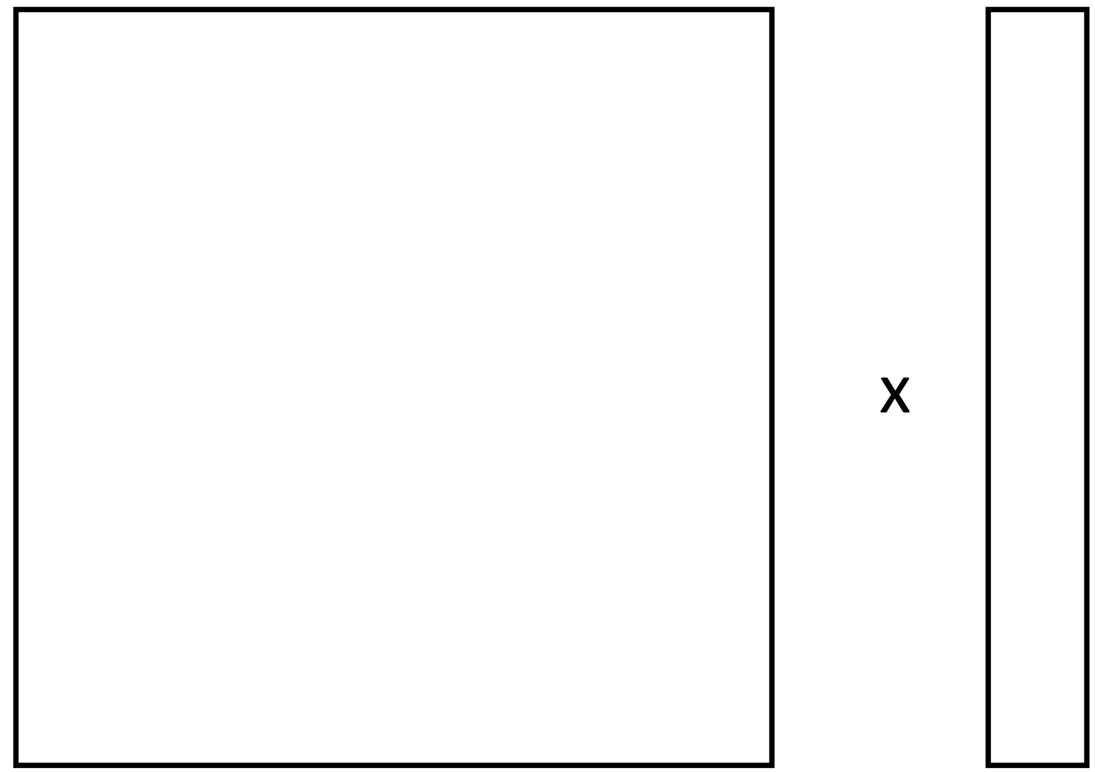
Matrix M
size m x n

Vector v
size n x 1

The matrix is stored in HDFS as a list of (i, j, m_{ij}) tuples

- The elements v_j of v are available to all mappers
- Map: ((i, j), m_{ij}) pair → ($i, m_{ij}v_j$) pair
- Reduce: (i, [$m_{i1}v_1, m_{i2}v_2, \ldots, m_{in}v_n$]) pair → ($i, m_{i1}v_1 + m_{i2}v_2 + \ldots + m_{in}v_n$) pair

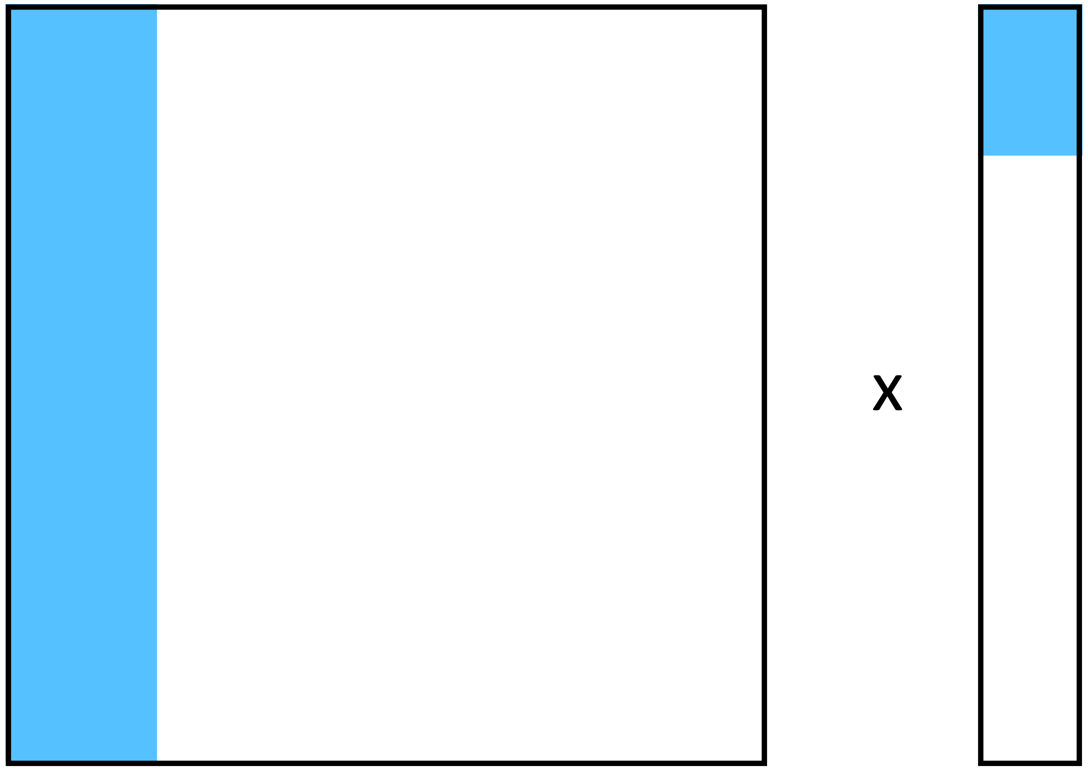
Vector does not fit



Matrix *M*
size *m* x *n*

Vector *v*
size *n* x *1*

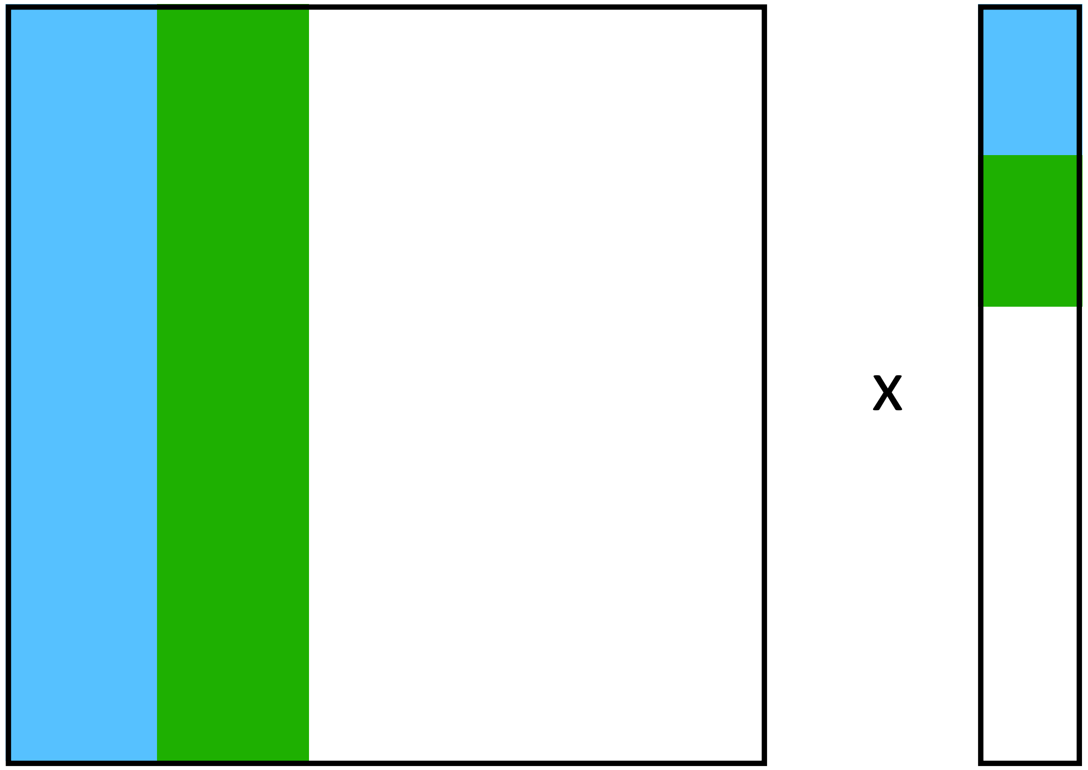
Vector does not fit



Matrix *M*
size *m* x *n*

Vector *v*
size *n* x *1*

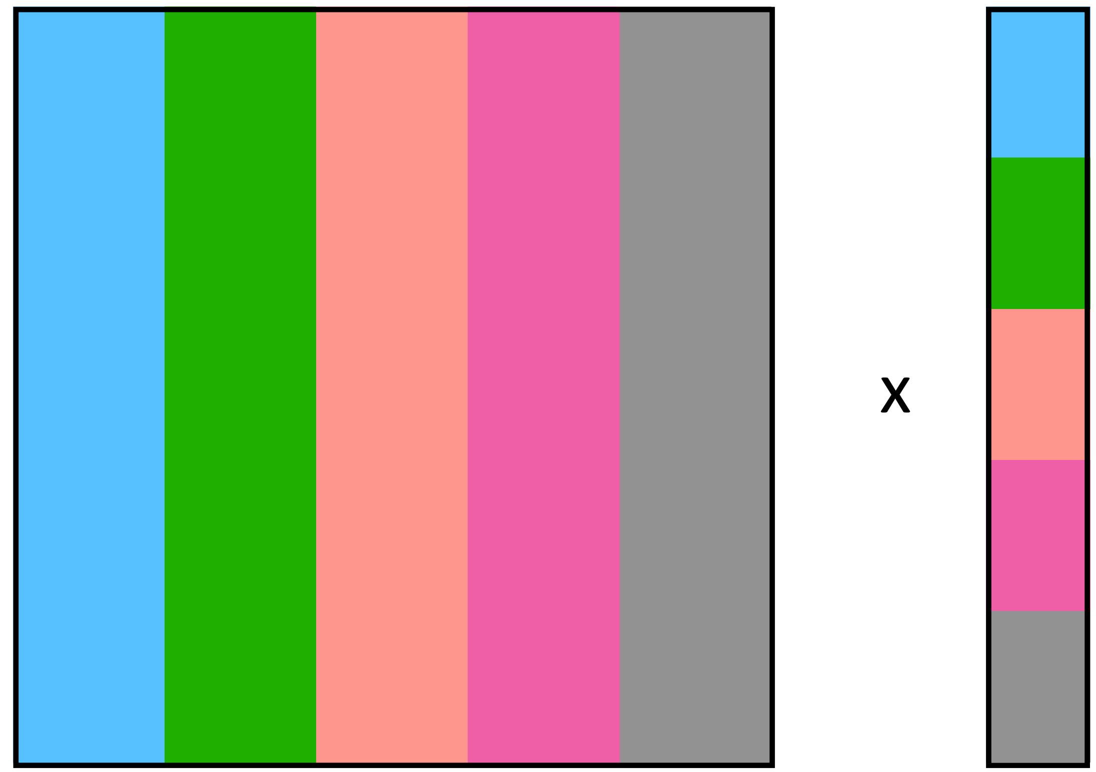
Vector does not fit



Matrix M
size m x n

Vector v
size n x 1

Vector does not fit

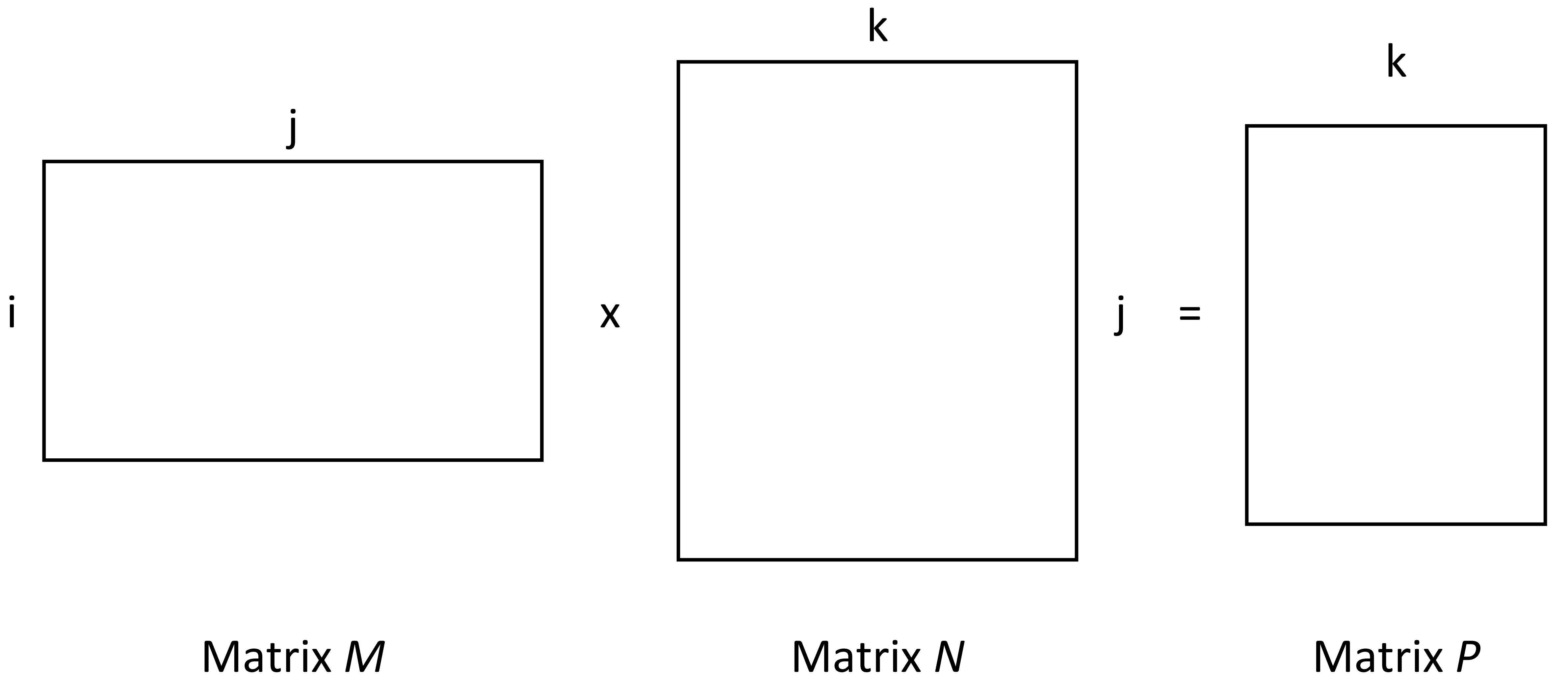


Matrix M
size m x n

Vector v
size n x 1

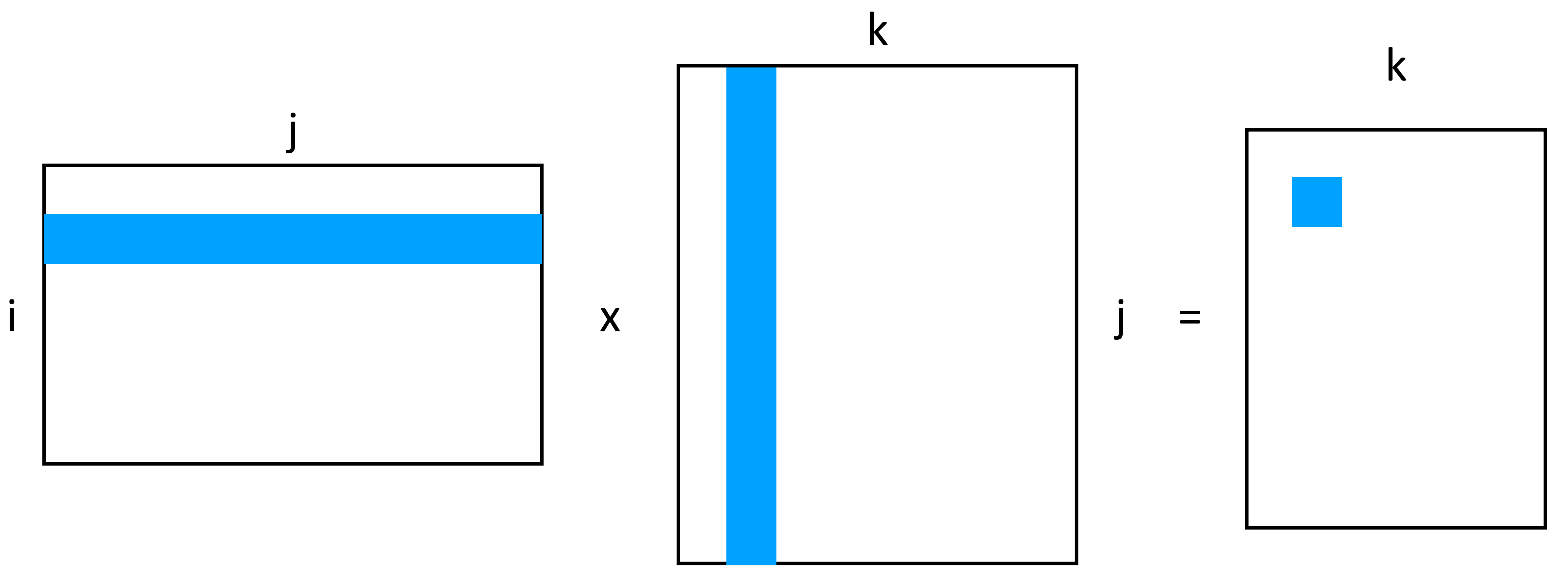
- Divide the vector in equal-sized subvectors that can fit in memory
- According to that, divide the matrix in stripes
- Stripe i and subvector i are independent from other stripes/subvectors
- Use the previous algorithm for each stripe/subvector pair

Matrix-Matrix Multiplication (I)



$$p_{ik} = \sum_j m_{ij} n_{jk}$$

Matrix-Matrix Multiplication (I)



Matrix *M* Matrix *N* Matrix *P*

$$p_{ik} = \sum_{j} m_{ij} n_{jk}$$

Matrix-Matrix Multiplication (II)

- A matrix can be seen as a 4-attributes relation/table:
  - (Matrix name, row index, column index, value) tuples
  - $M \rightarrow (M, i, j, m_{ij}), N \rightarrow (N, j, k, n_{jk})$
- As large matrices are often sparse (0's) we omit such tuples

Matrix-Matrix Multiplication (III)

Algorithm 1: The Map Function

for *each element* m_{ij} *of* M **do**
  produce $(key, value)$ pairs as $((i,k),(M,j,m_{ij}))$, for $k = 1,2,3,..$ up to the number of columns of N

for *each element* n_{jk} *of* N **do**
  produce $(key, value)$ pairs as $((i,k),(N,j,n_{jk}))$, for $i = 1,2,3,...$ up to the number of rows of M

return *Set of (key, value) pairs that each key,* (i,k)*, has a list with values* (M,j,m_{ij}) *and* (N,j,n_{jk}) *for all possible values of* j

Algorithm 2: The Reduce Function

for *each key (i,k)* **do**
  sort values begin with M by j in $list_M$
  sort values begin with N by j in $list_N$
  multiply m_{ij} and n_{jk} for j_{th} value of each list
  sum up $m_{ij} * n_{jk}$

return $(i,k), \sum_{j=1} m_{ij} * n_{jk}$

Matrix Matrix Multiplication (IV)

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} \longrightarrow \begin{bmatrix} 1a + 2c + 3e & 1b + 2d + 3f \\ 4a + 5c + 6e & 4b + 5d + 6f \end{bmatrix}$$

Matrix Matrix Multiplication (IV)

$$\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\end{bmatrix} \times \begin{bmatrix}a & b\\ c & d\\ e & f\end{bmatrix} \longrightarrow \begin{bmatrix}1a+2c+3e & 1b+2d+3f\\ 4a+5c+6e & 4b+5d+6f\end{bmatrix}$$

$(i,k),(M,j,m_{ij})$

$m_{11}=1$
$(1,1),(M,1,1)k=1$
$(1,2),(M,1,1)k=2$

$m_{12}=2$
$(1,1),(M,2,2)k=1$
$(1,2),(M,2,2)k=2$
....

$m_{23}=6$
$(2,1),(M,3,6)k=1$
$(2,2),(M,3,6)k=2$

$(i,k),(N,j,n_{jk})$

$n_{11}=a$
$(1,1),(N,1,a)i=1$
$(2,1),(N,1,a)i=2$

$n_{21}=c$
$(1,1),(N,2,c)i=1$
$(2,1),(N,2,c)i=2$

$n_{31}=e$
$(1,1),(N,3,e)i=1$
$(2,1),(N,3,e)i=2$
....

$n_{32}=f$
$(1,2),(N,3,f)i=1$
$(2,2),(N,3,f)i=2$

Matrix Matrix Multiplication (IV)

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} \longrightarrow \begin{bmatrix} 1a+2c+3e & 1b+2d+3f \\ 4a+5c+6e & 4b+5d+6f \end{bmatrix}$$

$$\begin{array}{l}
((i,k), [(M, j, m_{ij}), (M, j, m_{ij}), ..., (N, j, n_{jk}), (N, j, n_{jk}), ...]) \\
(1,1), [(M,1,1), (M,2,2), (M,3,3), (N,1,a, (N,2,c), (N,3,e)] \\
(1,2), [(M,1,1), (M,2,2), (M,3,3), (N,1,b, (N,2,d), (N,3,f)] \\
(2,1), [(M,1,4), (M,2,5), (M,3,6), (N,1,a, (N,2,c), (N,3,e)] \\
(2,2), [(M,1,4), (M,2,5), (M,3,6), (N,1,b, (N,2,d), (N,3,f)]
\end{array}$$

Matrix Matrix Multiplication (IV)

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} \qquad \begin{bmatrix} 1a + 2c + 3e & 1b + 2d + 3f \\ 4a + 5c + 6e & 4b + 5d + 6f \end{bmatrix}$$

$$[(M,1,1),(M,2,2),(M,3,3),(N,1,a,(N,2,c),(N,3,e)]$$

$$list_M = [(M,1,1),(M,2,2),(M,3,3)]$$
$$list_N = [(N,1,a),(N,2,c),(N,3,e)]$$

$$P(1,1) = 1a + 2c + 3e$$

$$P(1,1) = 1a + 2c + 3e$$
$$P(1,2) = 1b + 2d + 3f$$
$$P(2,1) = 4a + 5c + 6e$$
$$P(2,2) = 4b + 5d + 6f$$